

Result:
70%
7 / 10



Neptun ID:
Draft papers: ☐
Name: _____

The test consists of **3 questions**. The available time to solve the quiz is **20 minutes**. Please use a blue coloured pen; the use of any other tools is forbidden.

- 1.) (essay)
a b c d
2.) ☐☐☐☐
3.) ☐☐☐☐

Please copy the multiple choice questions' solution in the attached table. For each correct answer marked, 2 points are awarded; for each incorrect answer marked, -1 point is awarded. There is no limit to the number of correct answers per question. A negative aggregate score will be interpreted as 0 points.

1.) Define the semantic value of $(A \wedge B)$, e.g., by using a truth-table!

Score: 2 / 2.

2.) Which two interpretations are models of the following set of formulae?

$$\{ (((q \equiv p) \wedge p) \equiv q), \neg(q \vee q) \}$$

2 $\varrho(p) = 0, \quad \varrho(q) = 0$

(b) $\varrho(p) = 0, \quad \varrho(q) = 1$

2 $\varrho(p) = 1, \quad \varrho(q) = 0$

(d) $\varrho(p) = 1, \quad \varrho(q) = 1$

3.) Which two statements are true regarding the following formula?

$$\neg q \supset \neg p$$

2 a disjunctive normal form for the formula above is: $\neg p \vee q$

(b) a disjunctive normal form for the formula above is: $\neg p \wedge q$

0 a conjunctive normal form for the formula above is: $\neg p \vee q$

-1 a normal form for the formula above is: $\neg q \supset \neg p$

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